

# matrixdiagrams.sty

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## 1 Main command

`\matrixbrace[scale][aspect]{content}`

### 1.1 Scale (aspect = 1)

|                  |                                        |                  |                                        |
|------------------|----------------------------------------|------------------|----------------------------------------|
| $\left[ \right]$ | <code>\matrixbrace[1]{\mbBlank}</code> | $\left[ \right]$ | <code>\matrixbrace[2]{\mbBlank}</code> |
| $\left[ \right]$ | <code>\matrixbrace[3]{\mbBlank}</code> | $\left[ \right]$ | <code>\matrixbrace[4]{\mbBlank}</code> |
| $\left[ \right]$ | <code>\matrixbrace[5]{\mbBlank}</code> |                  |                                        |

## 1.2 Aspect ratio (scale = 3)

|                                                     |                                             |                                                     |                                           |
|-----------------------------------------------------|---------------------------------------------|-----------------------------------------------------|-------------------------------------------|
| $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | <code>\matrixbrace[3][0.5]{\mbBlank}</code> | $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | <code>\matrixbrace[3][1]{\mbBlank}</code> |
| $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | <code>\matrixbrace[3][1.5]{\mbBlank}</code> | $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | <code>\matrixbrace[3][2]{\mbBlank}</code> |
| $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | <code>\matrixbrace[3][3]{\mbBlank}</code>   |                                                     |                                           |

## 2 Diagonal commands

### 2.1 `\mbDiag` — main anti-diagonal

|                                                     |                                                     |                                          |
|-----------------------------------------------------|-----------------------------------------------------|------------------------------------------|
| $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | <code>\matrixbrace[3]{\mbDiag}</code>    |
|                                                     |                                                     | <code>\matrixbrace[3][2]{\mbDiag}</code> |

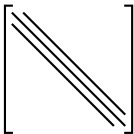
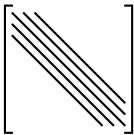
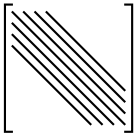
### 2.2 `\mbDiagBand` — tridiagonal band

Three parallel anti-diagonals with spacing  $d = 0.1$ .

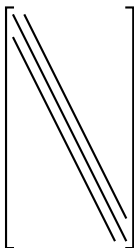
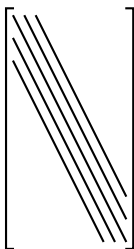
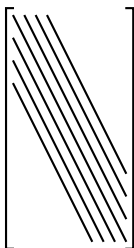
|                                                     |                                                     |                                              |
|-----------------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | $\left[ \begin{array}{c} \\ \\ \end{array} \right]$ | <code>\matrixbrace[3]{\mbDiagBand}</code>    |
|                                                     |                                                     | <code>\matrixbrace[3][2]{\mbDiagBand}</code> |

### 2.3 `\mbDiagBandN{n}` — variable band width

$n = 3, 5, 7, 9$  (spacing fixed at  $d = 0.1$  per diagonal):

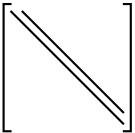
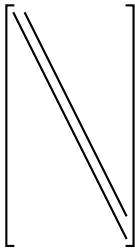
|                                                                                   |                                               |
|-----------------------------------------------------------------------------------|-----------------------------------------------|
|  | <code>\matrixbrace[3]{\mbDiagBandN{3}}</code> |
|  | <code>\matrixbrace[3]{\mbDiagBandN{5}}</code> |
|  | <code>\matrixbrace[3]{\mbDiagBandN{7}}</code> |
|  | <code>\matrixbrace[3]{\mbDiagBandN{9}}</code> |

Note: `\mbDiagBandN{3}` is identical to `\mbDiagBand`. With `aspect = 2`:

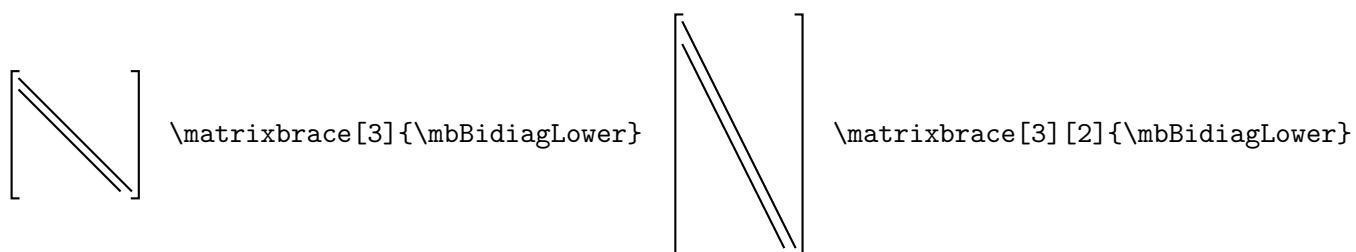
|                                                                                     |                                                  |
|-------------------------------------------------------------------------------------|--------------------------------------------------|
|   | <code>\matrixbrace[3][2]{\mbDiagBandN{3}}</code> |
|  | <code>\matrixbrace[3][2]{\mbDiagBandN{5}}</code> |
|  | <code>\matrixbrace[3][2]{\mbDiagBandN{7}}</code> |

### 3 Bidiagonal commands

#### 3.1 `\mbBidiagUpper` — main + super-diagonal

|                                                                                     |                                              |                                                                                     |                                                 |
|-------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------|
|  | <code>\matrixbrace[3]{\mbBidiagUpper}</code> |  | <code>\matrixbrace[3][2]{\mbBidiagUpper}</code> |
|-------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------|

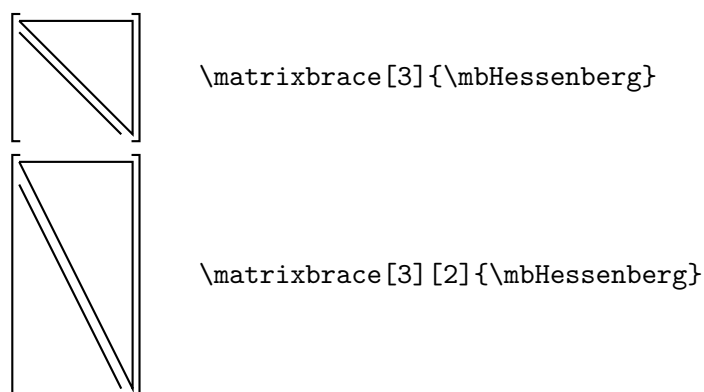
### 3.2 `\mbBidiagLower` — main + sub-diagonal



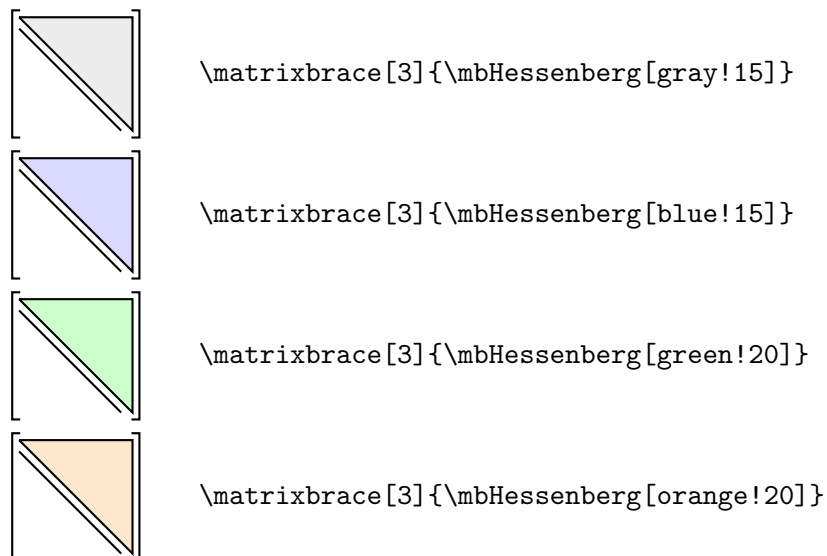
## 4 Hessenberg command

### 4.1 `\mbHessenberg` — outline only (default)

Upper-triangular outline  $(0, H) \rightarrow (1, H) \rightarrow (1, 0)$  plus one sub-diagonal.



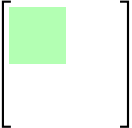
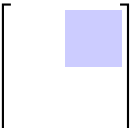
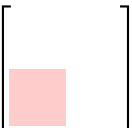
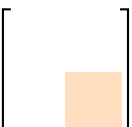
### 4.2 `\mbHessenberg[color]` — with fill



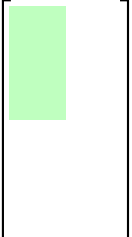
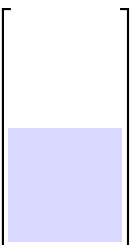
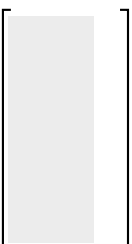
## 5 Fill and line commands

### 5.1 `\mbFill[color]{x0}{y0}{x1}{y1}`

Four quadrant fills; default color is `gray!20`:

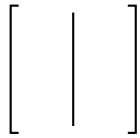
|                                                                                   |                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------|
|  | <code>\matrixbrace[3]{\mbFill[green!30]{0}{0.5}{0.5}{1}}</code>  |
|  | <code>\matrixbrace[3]{\mbFill[blue!20]{0.5}{0.5}{1}{1}}</code>   |
|  | <code>\matrixbrace[3]{\mbFill[red!20]{0}{0}{0.5}{0.5}}</code>    |
|  | <code>\matrixbrace[3]{\mbFill[orange!25]{0.5}{0}{1}{0.5}}</code> |
|  | <code>\matrixbrace[3]{\mbFill{0}{0}{1}{1}}</code>                |

With `aspect = 2` ( $y$ -coordinates run to 2):

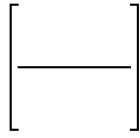
|                                                                                     |                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------|
|   | <code>\matrixbrace[3][2]{\mbFill[green!25]{0}{1}{0.5}{2}}</code> |
|  | <code>\matrixbrace[3][2]{\mbFill[blue!15]{0}{0}{1}{1}}</code>    |
|  | <code>\matrixbrace[3][2]{\mbFill[gray!15]{0}{0}{0.75}{2}}</code> |

## 5.2 `\mbVLine{x}` and `\mbHLine{y}`

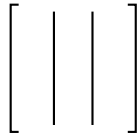
`\mbVLine{x}` runs from  $(x, 0)$  to  $(x, H)$ ; `\mbHLine{y}` runs from  $(0, y)$  to  $(1, y)$ .



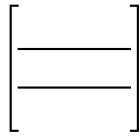
`\matrixbrace[3]{\mbVLine{0.5}}`



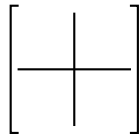
`\matrixbrace[3]{\mbHLine{0.5}}`



`\matrixbrace[3]{\mbVLine{0.33}\mbVLine{0.67}}`

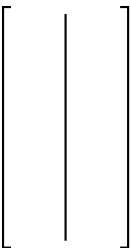


`\matrixbrace[3]{\mbHLine{0.33}\mbHLine{0.67}}`

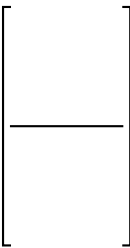


`\matrixbrace[3]{\mbVLine{0.5}\mbHLine{0.5}}`

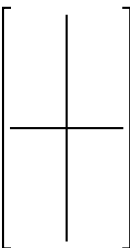
With `aspect = 2`:



`\matrixbrace[3][2]{\mbVLine{0.5}}`

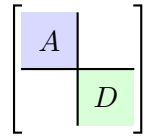
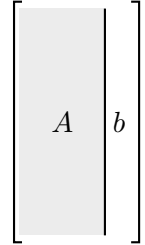


`\matrixbrace[3][2]{\mbHLine{1.0}}`



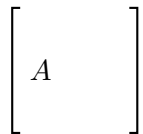
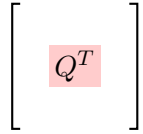
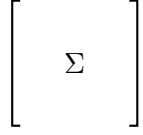
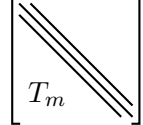
`\matrixbrace[3][2]{\mbVLine{0.5}\mbHLine{1.0}}`

Combined with fills (block partitioning):

|                                                                                   |                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>\matrixbrace[3]{   \mbFill[blue!15]{0}{0.5}{0.5}{1}   \mbFill[green!15]{0.5}{0}{1}{0.5}   \mbVLine{0.5}\mbHLine{0.5}   \mbLabel{0.25}{0.75}{A}   \mbLabel{0.75}{0.25}{D} }</pre> |
|  | <pre>\matrixbrace[3][2]{   \mbFill[gray!15]{0}{0}{0.75}{2}   \mbVLine{0.75}   \mbLabel{0.375}{1}{A}   \mbLabel{0.875}{1}{b} }</pre>                                                   |

## 6 Label commands

### 6.1 `\mbLabel{x}{y}{text}` — **math mode**

|                                                                                     |                                                                 |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|
|    | <pre>\matrixbrace[3]{\mbLabel{0.2}{0.5}{A}}</pre>               |
|   | <pre>\matrixbrace[3]{\mbLabel[red!20]{0.5}{0.5}{Q^T}}</pre>     |
|  | <pre>\matrixbrace[3]{\mbLabel{0.5}{0.5}{\Sigma}}</pre>          |
|  | <pre>\matrixbrace[3]{\mbDiagBand\mbLabel{0.25}{0.2}{T_m}}</pre> |

### 6.2 `\mbLabelCenter{text}` — **always centered**

Automatically placed at  $(0.5, H/2)$  regardless of aspect ratio:

|                                                 |                                                      |
|-------------------------------------------------|------------------------------------------------------|
| $\left[ \begin{array}{c} Q \end{array} \right]$ | <code>\matrixbrace[3][1]{\mbLabelCenter{Q}}</code>   |
| $\left[ \begin{array}{c} Q \end{array} \right]$ | <code>\matrixbrace[3][2]{\mbLabelCenter{Q}}</code>   |
| $\left[ \begin{array}{c} Q \end{array} \right]$ | <code>\matrixbrace[3][3]{\mbLabelCenter{Q}}</code>   |
| $\left[ \begin{array}{c} Q \end{array} \right]$ | <code>\matrixbrace[3][0.5]{\mbLabelCenter{Q}}</code> |

### 6.3 `\mbLabelRaw{x}{y}{text}` — raw text

|                                                            |                                                                |
|------------------------------------------------------------|----------------------------------------------------------------|
| $\left[ \begin{array}{c} \text{dense} \end{array} \right]$ | <code>\matrixbrace[3]{\mbLabelRaw{0.5}{0.5}{dense}}</code>     |
| $\left[ \begin{array}{c} \approx 0 \end{array} \right]$    | <code>\matrixbrace[3]{\mbLabelRaw{0.5}{0.5}{\approx 0}}</code> |

## 7 Spike commands

`\mbSpike{x}{y}{dir}` draws an L-shaped corner at  $(x, y)$ . The four directions connect to different pairs of edges:



$$\left[ \begin{array}{c} \text{H} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbSpike{0.5}{0.5}{t1}}$$

$$\left[ \begin{array}{c} \text{L} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbSpike{0.5}{0.5}{tr}}$$

$$\left[ \begin{array}{c} \text{H} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbSpike{0.5}{0.5}{bl}}$$

$$\left[ \begin{array}{c} \text{L} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbSpike{0.5}{0.5}{br}}$$

t1 (top+left), tr (top+right), bl (bottom+left), br (bottom+right).

With fills:

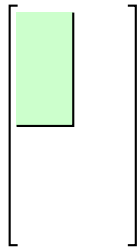
$$\left[ \begin{array}{c} \text{green box} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbFill[green!20]{0}{0.5}{0.5}{1}\mbSpike{0.5}{0.5}{t1}}$$

$$\left[ \begin{array}{c} \text{blue box} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbFill[blue!15]{0.5}{0.5}{1}{1}\mbSpike{0.5}{0.5}{tr}}$$

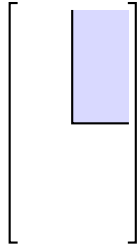
$$\left[ \begin{array}{c} \text{red box} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbFill[red!15]{0}{0}{0.5}{0.5}\mbSpike{0.5}{0.5}{bl}}$$

$$\left[ \begin{array}{c} \text{orange box} \\ \text{---} \end{array} \right] \quad \backslash\text{matrixbrace}[3]{\mbFill[orange!20]{0.5}{0}{1}{0.5}\mbSpike{0.5}{0.5}{br}}$$

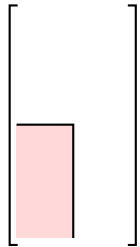
With aspect = 2, corner at mid-height (0.5, 1):



```
\matrixbrace[3][2]{\mbFill[green!20]{0}{1}{0.5}{2}\mbSpike{0.5}{1}{t1}}
```



```
\matrixbrace[3][2]{\mbFill[blue!15]{0.5}{1}{1}{2}\mbSpike{0.5}{1}{tr}}
```

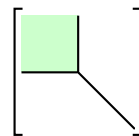


```
\matrixbrace[3][2]{\mbFill[red!15]{0}{0}{0.5}{1}\mbSpike{0.5}{1}{bl}}
```

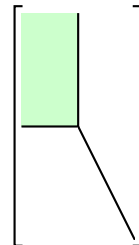
## 8 Clipped diagonal commands

The ...From variants force the diagonal band from  $[x_1, y_1]$  to  $[x_2, y_2]$ . Always pair with `\mbSpike` at the same corner.

### 8.1 `\mbDiagFrom{x1}{y1}{x2}{y2}`

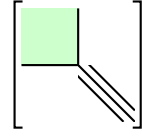
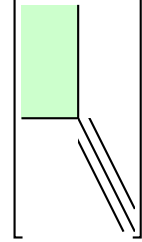


```
\matrixbrace[3]{
  \mbFill[green!20]{0}{0.5}{0.5}{1}
  \mbSpike{0.5}{0.5}{t1}
  \mbDiagFrom{0.5}{0.5}{1}{0}
}
```



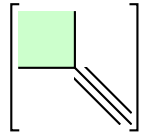
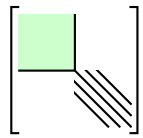
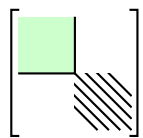
```
\matrixbrace[3][2]{
  \mbFill[green!20]{0}{1}{0.5}{2}
  \mbSpike{0.5}{1}{t1}
  \mbDiagFrom{0.5}{1}{1}{0}
}
```

## 8.2 `\mbDiagBandFrom{cx}{cy}`

|                                                                                   |                                                                                                                              |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>\matrixbrace[3]{   \mbFill[green!20]{0}{0.5}{0.5}{1}   \mbSpike{0.5}{0.5}{t1}   \mbDiagBandFrom{0.5}{0.5}{1}{0} }</pre> |
|  | <pre>\matrixbrace[3][2]{   \mbFill[green!20]{0}{1}{0.5}{2}   \mbSpike{0.5}{1}{t1}   \mbDiagBandFrom{0.5}{1}{1}{0} }</pre>    |

## 8.3 `\mbDiagBandNFrom{n}{cx}{cy}`

$n = 3, 5, 7$ :

|                                                                                     |                                                                                                                                  |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|    | <pre>\matrixbrace[3]{   \mbFill[green!20]{0}{0.5}{0.5}{1}   \mbSpike{0.5}{0.5}{t1}   \mbDiagBandNFrom{3}{0.5}{0.5}{1}{0} }</pre> |
|  | <pre>\matrixbrace[3]{   \mbFill[green!20]{0}{0.5}{0.5}{1}   \mbSpike{0.5}{0.5}{t1}   \mbDiagBandNFrom{5}{0.5}{0.5}{1}{0} }</pre> |
|  | <pre>\matrixbrace[3]{   \mbFill[green!20]{0}{0.5}{0.5}{1}   \mbSpike{0.5}{0.5}{t1}   \mbDiagBandNFrom{7}{0.5}{0.5}{1}{0} }</pre> |

# 9 Realistic equations

## 9.1 QR decomposition: $A = QR$

|                                                                                                     |                                                                                                                             |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} Q \end{bmatrix} \begin{bmatrix} R \end{bmatrix}$ | <pre>\matrixbrace[3][2]{\mbLabelCenter{A}} = \matrixbrace[3][2]{\mbLabelCenter{Q}} \matrixbrace[3]{\mbLabelCenter{R}}</pre> |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|

## 9.2 Bidiagonalization: $A = UBV^T$

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} U \end{bmatrix} \begin{bmatrix} \diagdown \diagup \end{bmatrix} \begin{bmatrix} V^T \end{bmatrix}$$

```
\matrixbrace[3]{\mbLabelCenter{A}} =
\matrixbrace[3]{\mbLabelCenter{U}}
\matrixbrace[3]{\mbBidiagUpper}
\matrixbrace[3]{\mbLabelCenter{V^T}}
```

## 9.3 Similarity: $Q^T T_m Q = \tilde{T}_m$

$$\begin{bmatrix} Q^T \end{bmatrix} \begin{bmatrix} \diagdown \diagup \end{bmatrix} \begin{bmatrix} Q \end{bmatrix} = \begin{bmatrix} \text{green square} \end{bmatrix} \begin{bmatrix} \tilde{T}_m \end{bmatrix}$$

```
\matrixbrace[3]{\mbLabelCenter{Q^T}}
\matrixbrace[3]{
\mbDiagBand\mbLabel{0.25}{0.15}{T_m}
}
\matrixbrace[3]{\mbLabelCenter{Q}} =
\matrixbrace[3]{
\mbFill[green!25]{0}{0.5}{0.5}{1}
\mbSpike{0.5}{0.5}{t1}
\mbDiagBandFrom{0.5}{0.5}{1}{0}
\mbLabel{0.3}{0.12}{\widetilde{T}_m}
}
```

## 9.4 Hessenberg reduction: $Q^T A Q = H$

$$\begin{bmatrix} Q^T \end{bmatrix} \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} Q \end{bmatrix} = \begin{bmatrix} \diagdown \diagup \end{bmatrix} \begin{bmatrix} H \end{bmatrix}$$

```
\matrixbrace[3]{\mbLabelCenter{Q^T}}
\matrixbrace[3]{\mbLabelCenter{A}}
\matrixbrace[3]{\mbLabelCenter{Q}} =
\matrixbrace[3]{
\mbHessenberg\mbLabel{0.7}{0.55}{H}
}
```

## 9.5 Column-bordered decomposition: $AP_m = P_m T_m + f$

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} P_m \end{bmatrix} = \begin{bmatrix} P_m \end{bmatrix} \begin{bmatrix} \diagdown \diagup \end{bmatrix} + \begin{bmatrix} \text{gray square} \end{bmatrix} \begin{bmatrix} f \end{bmatrix}$$

```
\matrixbrace[4]{\mbLabelCenter{A}}
\matrixbrace[2]{\mbLabelCenter{P_m}} =
\matrixbrace[2]{\mbLabelCenter{P_m}}
\matrixbrace[2]{
\mbDiagBand \mbLabel{0.25}{0.15}{T_m}
} +
\matrixbrace[2]{
\mbFill[gray!15]{0}{0}{0.75}{1}
\mbVLine{0.75} \mbLabel{0.875}{0.5}{f}
}
```

## 9.6 Schur complement structure

$$\begin{bmatrix} A & B \\ C & S \end{bmatrix}$$

```
\matrixbrace[4]{
\mbFill[blue!10]{0}{0.5}{0.5}{1} \mbFill[green!10]{0.5}{0}{1}{0.5}
\mbFill[gray!10]{0.5}{0.5}{1}{1}
\mbVLine{0.5} \mbHLine{0.5}
\mbLabel{0.25}{0.75}{A} \mbLabel{0.75}{0.75}{B}
\mbLabel{0.25}{0.25}{C} \mbLabel{0.75}{0.25}{S}}
```

## 9.7 Bordered similarity: $\overline{Q}_k^T D_{k+1} \overline{Q}_k = \overline{T}_{k+1}$

$$\begin{bmatrix} \overline{Q}_k^T \\ 1 \end{bmatrix} \begin{bmatrix} D_{k+1} \\ \hline \hline \end{bmatrix} \begin{bmatrix} \overline{Q}_k \\ 1 \end{bmatrix} = \begin{bmatrix} \overline{T}_{k+1} \end{bmatrix}$$

```
\matrixbrace[3]{
\mbFill[magenta!10]{0}{0.15}{0.85}{1}
\mbLabel{0.5}{0.5}{\overline{Q}_{k}^T}
\mbLabel{0.93}{0.08}{1}
}
\matrixbrace[3]{
\mbDiag
\draw[thick] (0,0) -- (1,0) -- (1,\mbH);
\mbLabel{0.25}{0.45}{D_{k+1}}
}
\matrixbrace[3]{
\mbFill[magenta!10]{0}{0.15}{0.85}{1}
\mbLabel{0.5}{0.5}{\overline{Q}_{k}}
\mbLabel{0.93}{0.08}{1}
} =
\matrixbrace[3]{
\mbDiagBand
\mbLabel{0.25}{0.15}{\overline{T}_{k+1}}
}
```

## 10 Command reference

|                                                  |                                                                                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------|
| <code>\matrixbrace[s][a]{c}</code>               | Main command. <code>s</code> = scale (def. 1), <code>a</code> = aspect (def. 1 = square). |
| <code>\mbDiag</code>                             | Anti-diagonal $(0, H) \rightarrow (1, 0)$ .                                               |
| <code>\mbDiagBand</code>                         | Tridiagonal band, spacing $d = 0.1$ .                                                     |
| <code>\mbDiagBandN{n}</code>                     | $n$ -diagonal band ( $n$ odd), fixed step 0.1.                                            |
| <code>\mbBidiagUpper</code>                      | Main diagonal + super-diagonal ( $d = 0.1$ ).                                             |
| <code>\mbBidiagLower</code>                      | Main diagonal + sub-diagonal ( $d = 0.1$ ).                                               |
| <code>\mbHessenberg</code>                       | Upper-triangular outline + sub-diagonal.                                                  |
| <code>\mbHessenberg[color]</code>                | Same with filled interior.                                                                |
| <code>\mbDiagFrom{x1}{y1}{x2}{y2}</code>         | Main diagonal, from $(x_1, y_1)$ to $(x_2, y_2)$                                          |
| <code>\mbDiagBandFrom{x1}{y1}{x2}{y2}</code>     | Tridiagonal band, clipped.                                                                |
| <code>\mbDiagBandNFrom{n}{x1}{y1}{x2}{y2}</code> | $n$ -diagonal band, clipped.                                                              |
| <code>\mbFill[col]{x0}{y0}{x1}{y1}</code>        | Filled rectangle. Default color <code>gray!20</code> .                                    |
| <code>\mbVLine{x}</code>                         | Vertical line, full height 0 to $H$ .                                                     |
| <code>\mbHLine{y}</code>                         | Horizontal line, full width.                                                              |
| <code>\mbLabel{x}{y}{text}</code>                | Math-mode label at $(x, y)$ .                                                             |
| <code>\mbLabel[color]{x}{y}{text}</code>         | Same with filled node label                                                               |
| <code>\mbLabelRaw{x}{y}{text}</code>             | Raw-text label at $(x, y)$ .                                                              |
| <code>\mbLabelCenter{text}</code>                | Math label at center $(0.5, H/2)$ , any aspect.                                           |
| <code>\mbSpike{x}{y}{dir}</code>                 | L-corner; <code>dir</code> $\in \{\text{tl}, \text{tr}, \text{bl}, \text{br}\}$ .         |
| <code>\mbBlank</code>                            | Empty content (brackets only).                                                            |
| <code>\mbH</code>                                | Expands to current aspect value inside content.                                           |